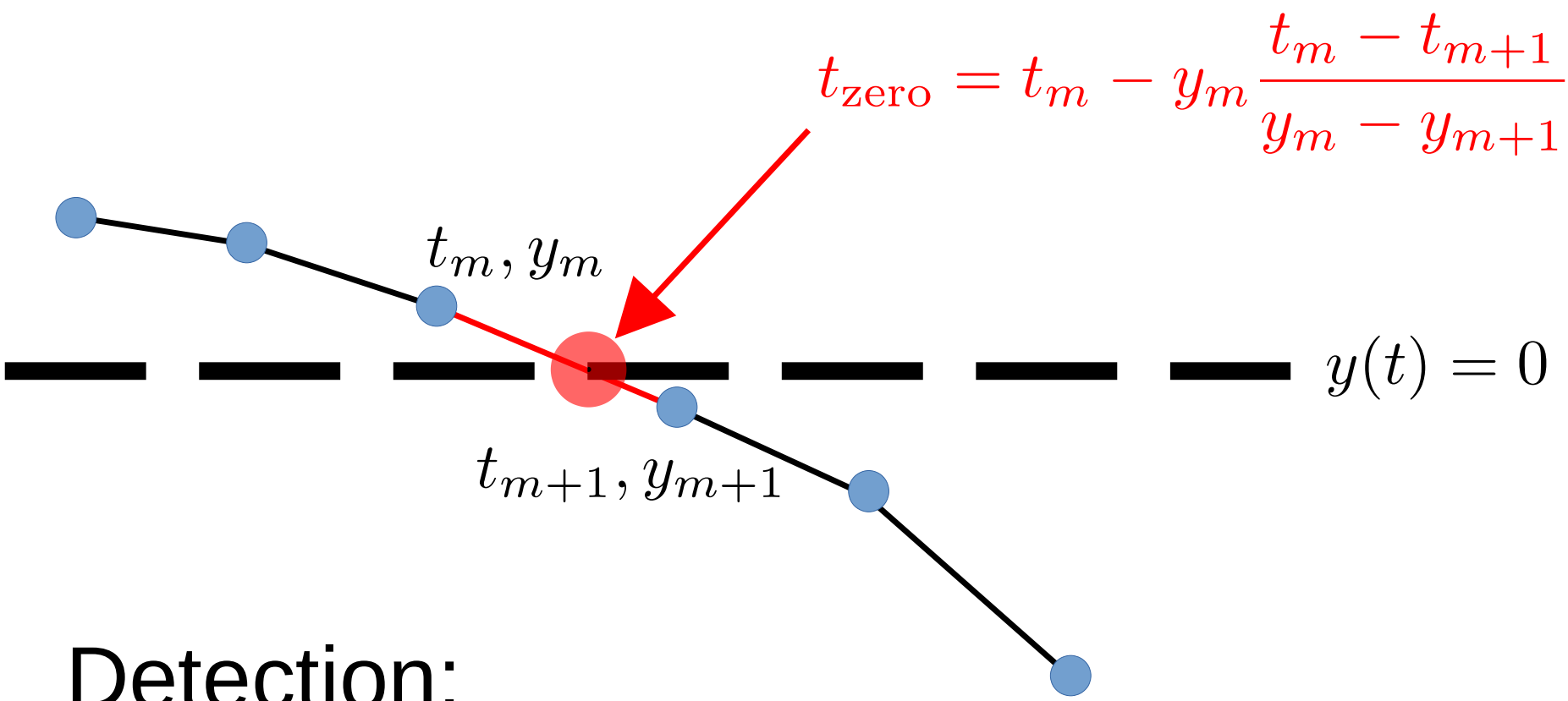

Detection:

$y_m y_{m+1} < 0$